

Blockchain and Cryptocurrencies Technologies

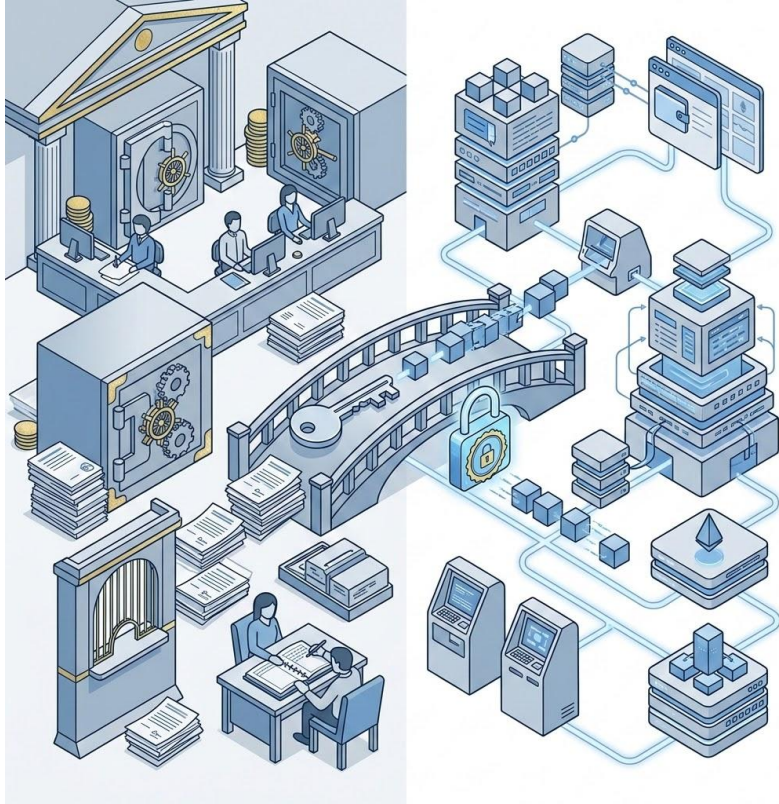
DeFi and On Chain Finance
Smart Contracts, Liquidity, Collateral, and Risk

DeFi and On Chain Finance

- Stablecoins make money programmable
- DeFi makes financial functions programmable
- This lecture studies the financial machines built around tokens, stablecoins, wallets, smart contracts, liquidity, and collateral
- The goal is not to list DeFi protocols
- The goal is to understand the basic machinery behind decentralized financial systems
- DeFi is not finance without intermediaries
- It is finance where intermediation moves into code, liquidity, oracles, governance, and interfaces



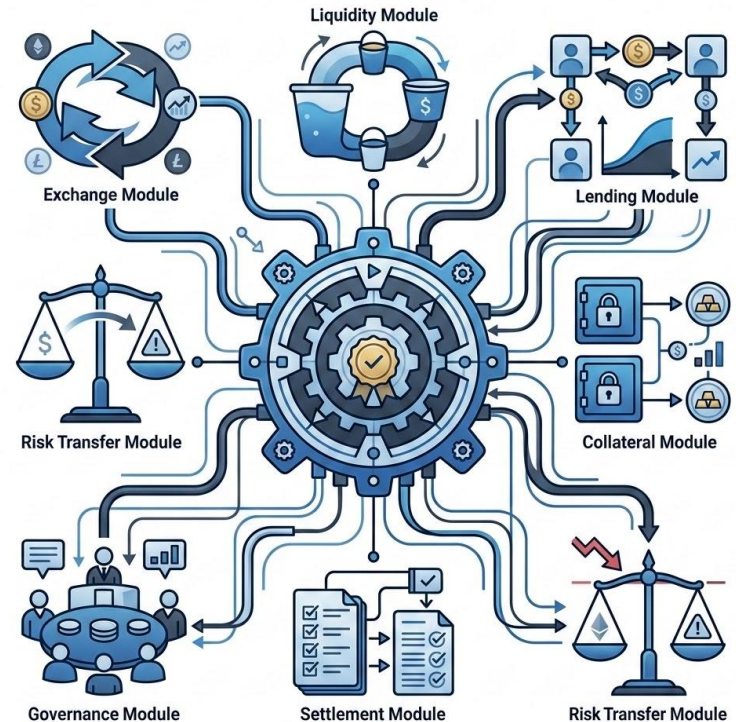
From Financial Institutions to Financial Protocols



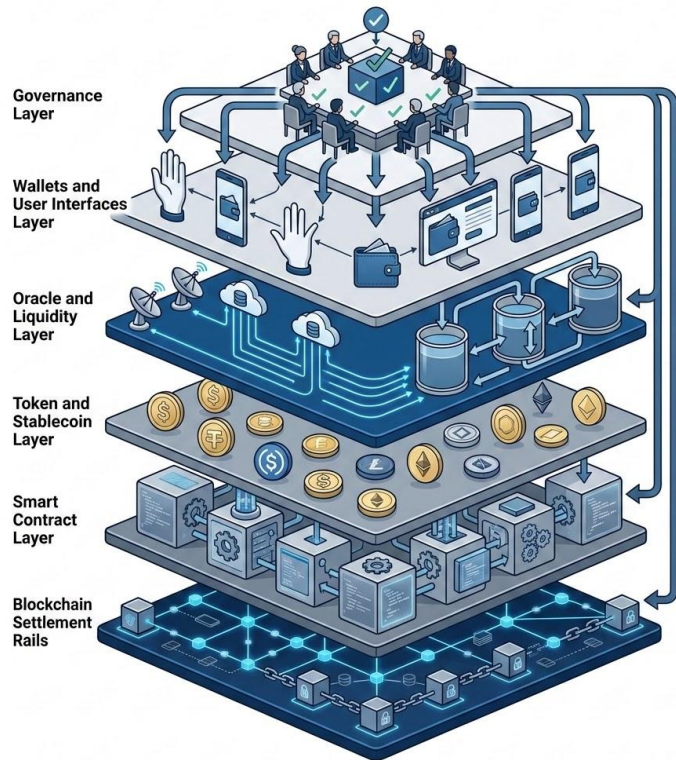
- Traditional finance depends on banks, brokers, custodians, clearing houses, contracts, ledgers, and trusted intermediaries
- DeFi replaces some of these functions with smart contracts and public settlement
- The user interacts through wallets and interfaces
- The protocol enforces rules automatically
- Intermediation does not disappear
- It changes form
- The bank counter disappears, but protocol risk appears

What DeFi Actually Automates

- DeFi automates financial workflows that used to require institutions
- Exchange
- Liquidity provision
- Borrowing and lending
- Collateral management
- Interest rate adjustment
- Liquidation
- Settlement
- Governance rules
- Risk transfer
- DeFi automates workflows, not trust itself



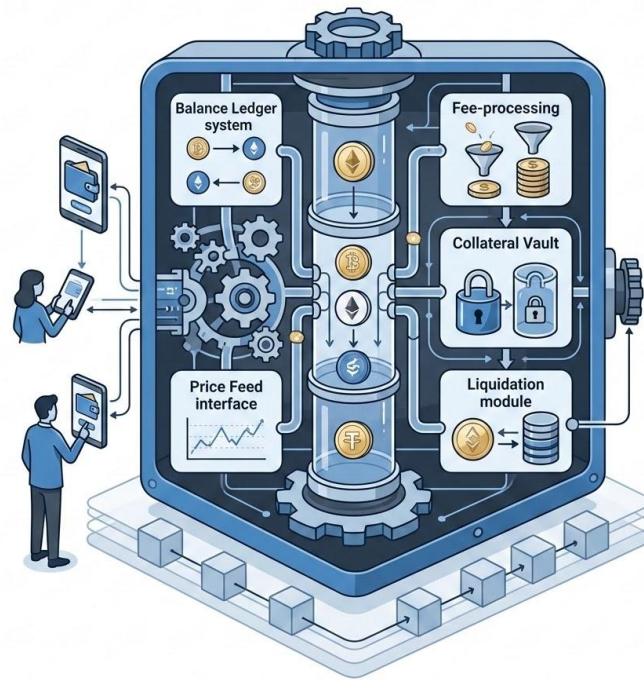
The DeFi Stack



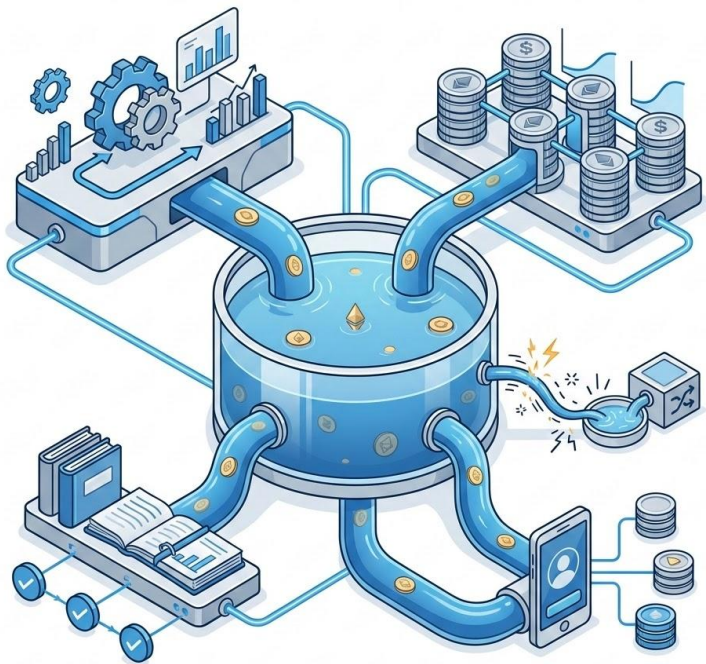
- DeFi is not one protocol
- It is a stack of dependencies
- Settlement layer: blockchain where transactions are finalized
- Smart contracts: programmable rules that hold and move assets
- Tokens and stablecoins: assets used inside protocols
- Oracles: external data entering on chain logic
- Liquidity: assets available for trading, lending, and settlement
- Wallets and interfaces: how users interact with protocols
- Governance: how parameters and upgrades are decided

Smart Contracts as Financial State Machines

- Smart contracts hold assets
- They enforce financial rules
- They execute deterministically
- They calculate balances, fees, prices, collateral ratios, and liquidation conditions
- They replace discretionary back office processes with programmable logic
- They are transparent, but not automatically safe
- A bug becomes part of the financial system
- In DeFi, the contract is not paperwork
- The contract is the machine



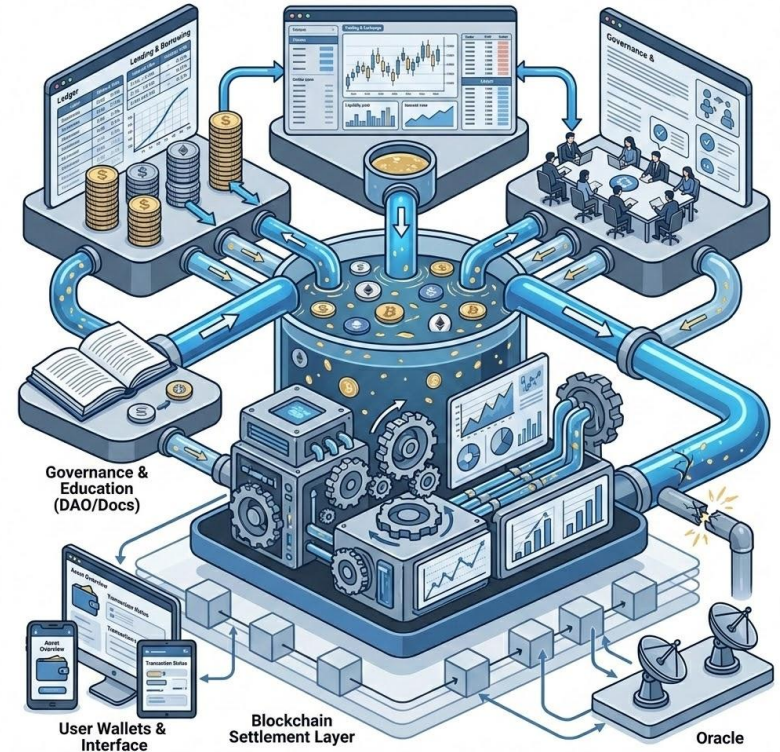
Liquidity Is the Fuel of DeFi



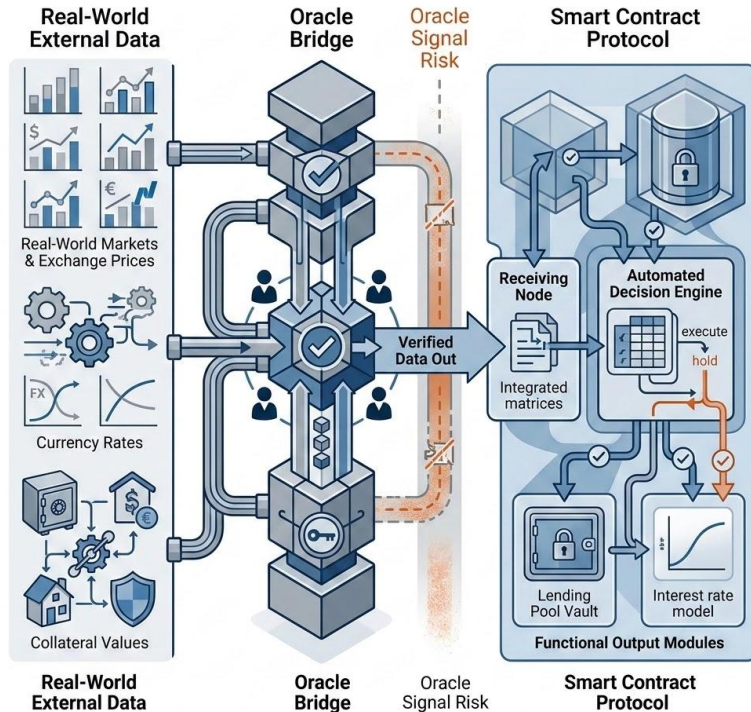
- Liquidity means the ability to buy, sell, borrow, lend, or settle an asset with low price impact and low friction
- A liquid market allows large transactions without dramatically changing the price
- A liquid protocol has enough assets available for users to trade, borrow, repay, redeem, or exit
- DeFi protocols need liquidity before they can be useful
- Liquidity reduces slippage, improves confidence, and attracts integrations
- Without liquidity, a protocol may work technically but fail economically
- No liquidity, no DeFi
- Just code waiting for money

TVL: Useful Metric, Dangerous Shortcut

- TVL means Total Value Locked
- It measures the total value of assets deposited in a protocol or set of protocols
- TVL can indicate how much capital users trust to a protocol
- But TVL is not trading volume, revenue, profit, or real adoption
- TVL can be inflated by incentives, recursive deposits, leverage loops, and double counting
- High TVL does not prove security
- High TVL does not prove meaningful usage
- Use TVL together with volume, active users, fees, revenue, liquidity depth, and risk
- TVL tells you how much money is inside
- It does not tell you whether the machine is healthy



Oracles: The Outside World Enters DeFi



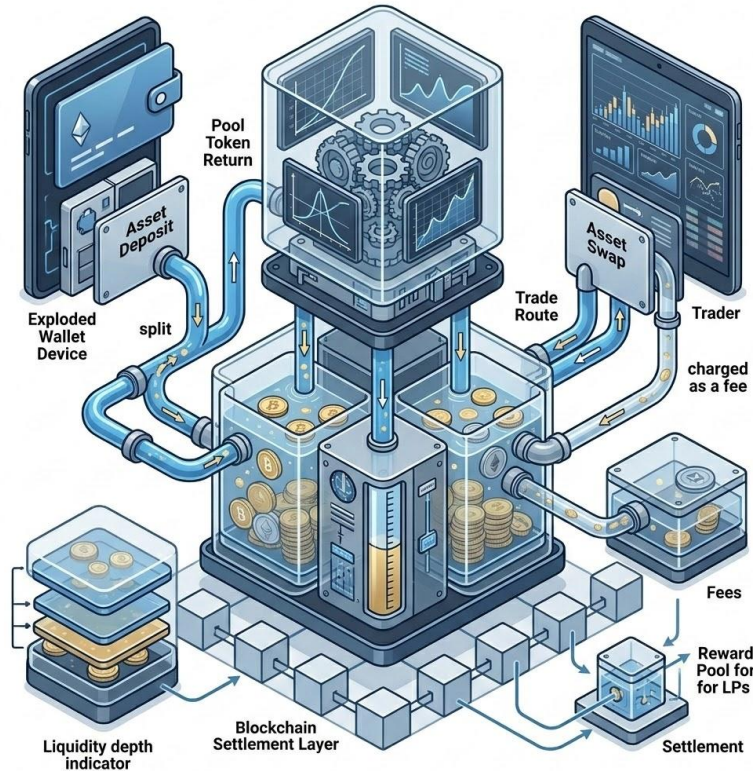
- Smart contracts cannot naturally know external facts
- They do not know the ETH price, the BRL/USD rate, the value of collateral, or whether an off chain asset exists
- Oracles bring external data into smart contracts
- DeFi lending depends heavily on price oracles
- Liquidations depend on oracle values
- Stablecoins, derivatives, and risk engines often need external data
- Oracle failure can become protocol failure
- DeFi is only as decentralized as the data it depends on

Decentralized Exchanges

- A decentralized exchange is a protocol that allows users to swap tokens through smart contracts
- Users trade from their wallets
- The protocol executes trades without a traditional centralized exchange custody model
- Some DEXs use order books
- Many major DeFi DEXs use Automated Market Makers
- DEXs reduce dependence on centralized trading platforms
- But they introduce smart contract risk, liquidity risk, slippage, MEV exposure, and interface risk
- A DEX moves exchange logic from a company server to a protocol



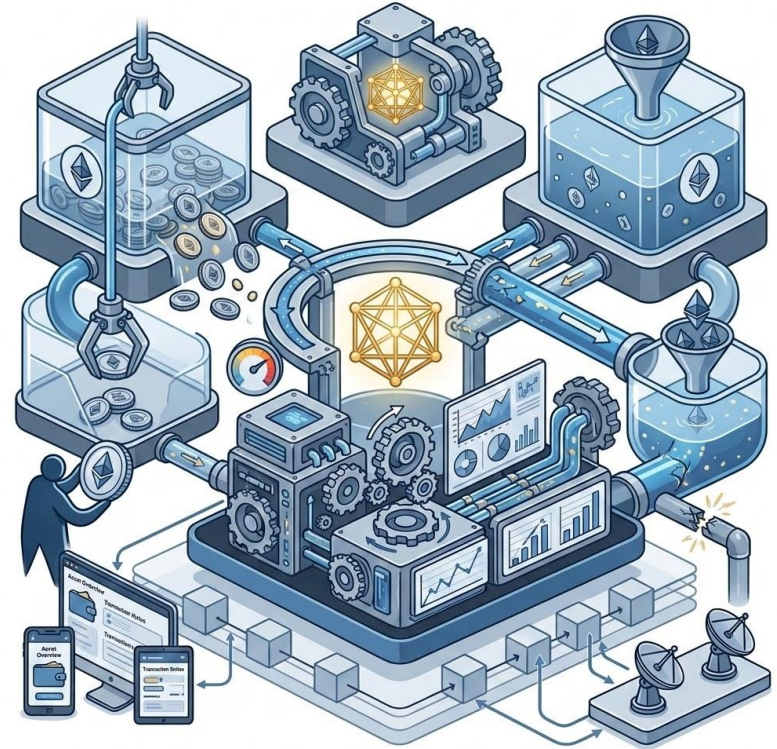
Automated Market Makers and Liquidity Pools



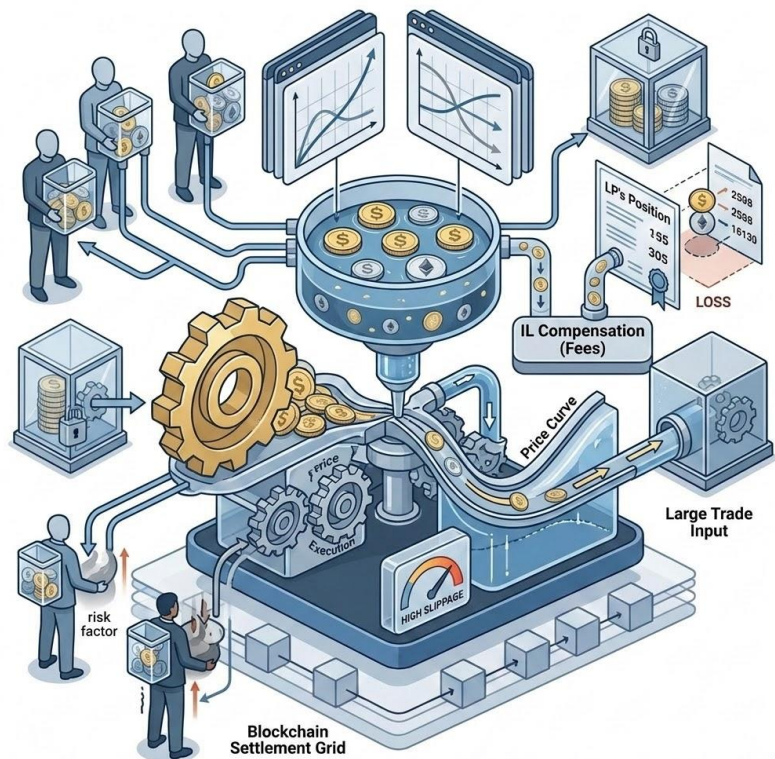
- An Automated Market Maker is a smart contract system that prices trades using a formula and a liquidity pool
- A liquidity pool is a smart contract that holds assets available for trading or other protocol operations
- Instead of matching buyers and sellers directly, traders swap against pooled assets
- Liquidity providers deposit assets into the pool
- The AMM updates prices automatically as pool balances change
- The pool charges fees, which may be distributed to liquidity providers
- The depth of the pool determines how much trading it can absorb
- An AMM is not matching buyers and sellers
- It lets traders buy from a pool

Constant Product Invariant

- Many AMMs use the constant product invariant
- The basic intuition is x times y equals k
 - x is the amount of one asset in the pool
 - y is the amount of the other asset in the pool
 - k is the invariant the pool tries to preserve after each trade, ignoring fees
- When a trader removes one asset from the pool, they must add enough of the other asset to preserve the invariant
- The asset being bought becomes more expensive as it becomes scarcer in the pool
- The formula creates automatic pricing, but not infinite liquidity
- The formula prices liquidity
- It does not create infinite liquidity



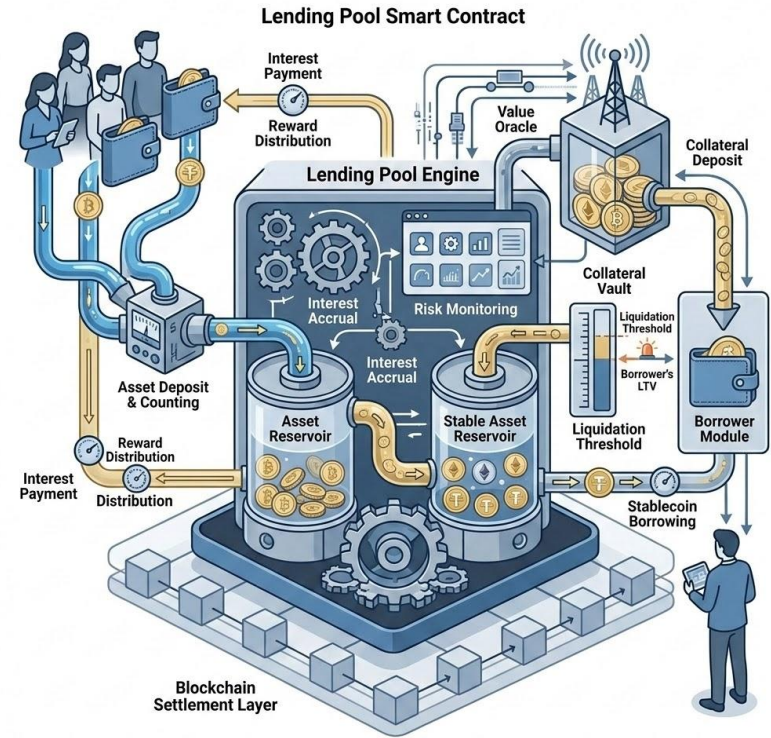
Slippage and Impermanent Loss



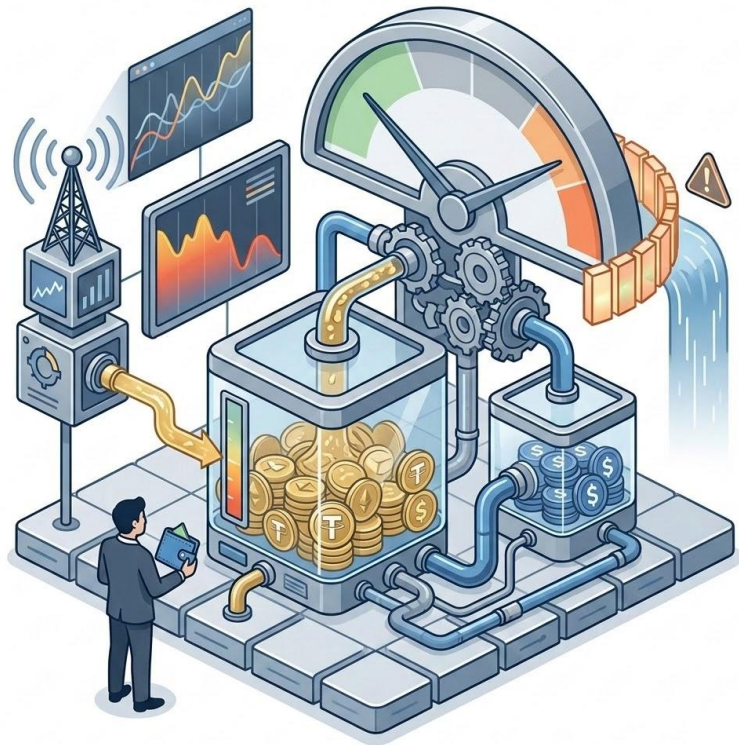
- Slippage is the difference between the expected price of a trade and the actual executed price
- In AMMs, slippage happens because the trade changes pool balances while it is executed
- Large trades create more slippage
- Shallow pools create more slippage
- Impermanent loss affects liquidity providers when relative asset prices change after deposit
- The LP may end up with less value than if they had simply held the assets
- Fees may compensate, but not always
- Providing liquidity is not passive risk free income
- Slippage is what the market charges you for being too large for the pool
- Liquidity providers are taking inventory risk

Lending and Borrowing Protocols

- DeFi lending protocols allow users to deposit assets into pools
- Other users borrow assets from those pools
- Loans are usually overcollateralized
- The borrower deposits collateral worth more than the borrowed amount
- Interest rates often adjust according to utilization
- Stablecoins are common borrowed assets because debt in stable value is easier to reason about
- DeFi lending replaces credit assessment with collateral management
- In DeFi, the protocol does not ask who you are
- It asks what collateral you posted



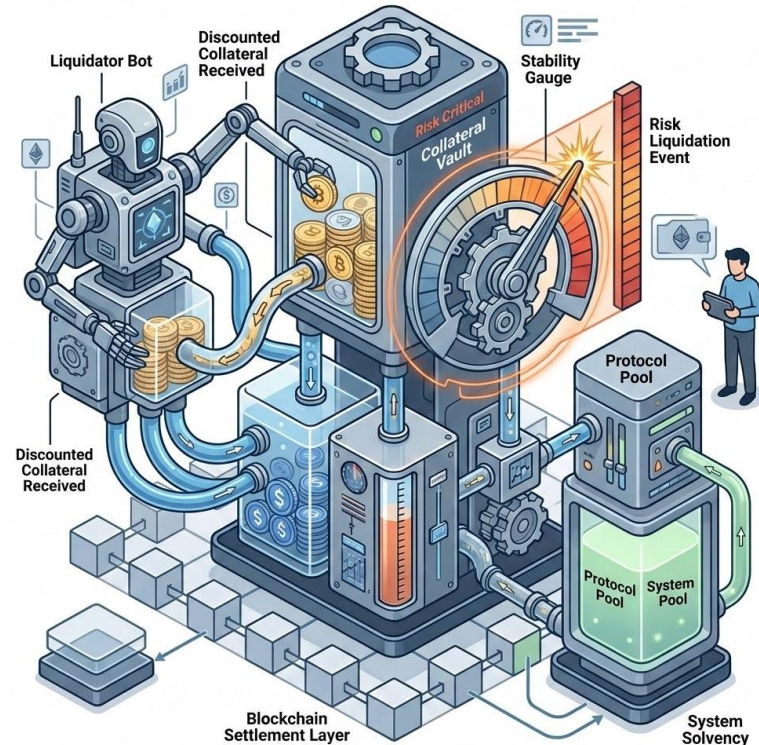
Collateral and Health Factor



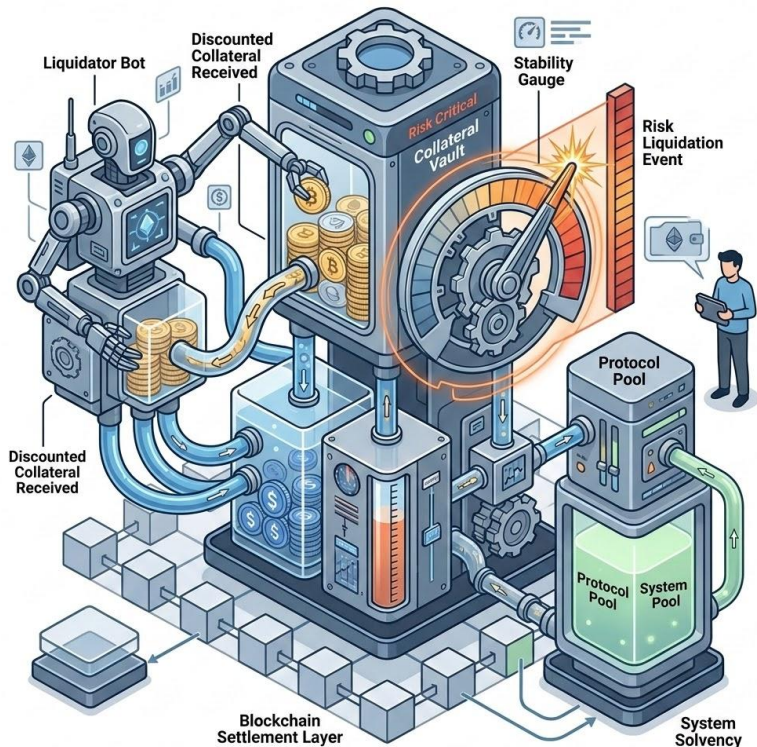
- The protocol tracks collateral value against borrowed value
- A safe position has more collateral value than debt value
- Health factor is a safety indicator for a borrowing position
- If collateral value falls or borrowed value rises, the health factor worsens
- When the health factor crosses a threshold, liquidation can happen
- Oracles are critical because they provide the prices used to calculate collateral value
- In DeFi, your credit score is your collateral ratio
- Example: depositing volatile collateral to borrow stablecoins looks safe only while collateral value remains high enough

Liquidations

- Liquidation protects the solvency of the protocol
- When a borrower becomes undercollateralized, liquidators can repay part of the debt
- In exchange, liquidators receive collateral at a discount
- This creates an incentive for external actors or bots to keep the protocol solvent
- Liquidation is automatic and rule based
- It can be brutal for users during volatility
- Oracle delays, congestion, or manipulation can make liquidations dangerous
- Liquidation is the protocol protecting itself from your optimism



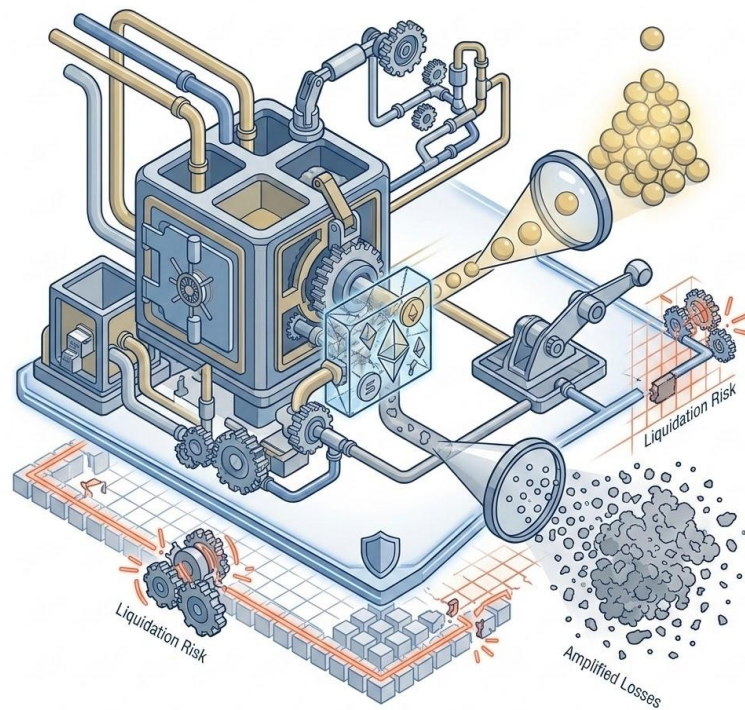
Leverage Pools



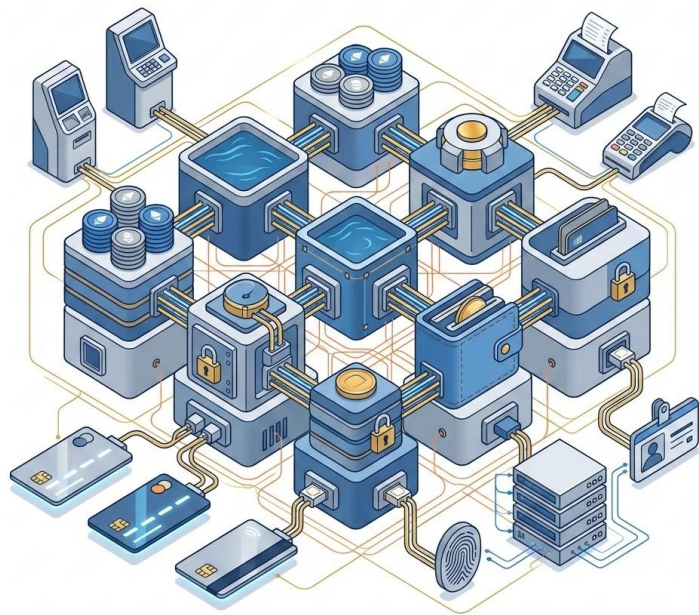
- A leverage pool is a pooled structure that creates leveraged exposure to an asset, strategy, or liquidity position
 - Users deposit capital into the pool
 - The pool may borrow additional assets or use derivatives to increase exposure
- Leverage can amplify returns when the strategy works
- Leverage also amplifies losses when prices move against the position
 - In DeFi, leverage pools can be used for leveraged trading, leveraged liquidity provision, or structured yield strategies
- They depend on collateral rules, liquidation thresholds, liquidity depth, and oracle accuracy
- Small price movements can become solvency problems
- Leverage pools do not create free yield

Stablecoins Inside Lending

- Stablecoins are common debt assets in DeFi lending
- Borrowers often borrow stablecoins against volatile collateral
- This creates predictable debt value
- Stablecoins can also be used as collateral in some systems
- A stablecoin depeg can damage lending markets
- If the stablecoin used as collateral loses value, lenders may become exposed
- If the stablecoin used as debt depegs, borrowers and lenders face unexpected incentives
- Stablecoin risk becomes lending risk
- Stablecoin risk does not disappear inside DeFi
- It becomes collateral risk, debt risk, and liquidity risk



Composability and Hybrid Finance



- Composability means protocols can plug into each other
- A token from one protocol can be used in another protocol
- Liquidity pool tokens, stablecoins, collateral tokens, receipt tokens, and wrapped assets can become building blocks
- Composability accelerates innovation
- It also accelerates contagion
- Not everything with tokens is DeFi
- Many real products mix smart contracts, centralized interfaces, custodians, KYC, cards, PIX, exchanges, and banking rails
- Composability is DeFi's superpower and its attack surface
- The future is not pure DeFi
- It is messy hybrid finance

The Main Risks of DeFi

- Smart contract bugs can lock or drain assets
- Oracle manipulation can distort prices
- Liquidity crises can make exits expensive or impossible
- Bridge failures can break wrapped or cross chain assets
- Governance attacks can change protocol parameters
- Leverage loops can amplify losses
- MEV means Maximal Extractable Value: value extracted by influencing transaction ordering
- DEX trades can be front run or sandwiched
- Interfaces can hide real protocol risk
- Regulation can return through ramps, front ends, stablecoins, custodians, and identity layers
- DeFi does not eliminate financial risk
- It makes risk faster, programmable, and composable

